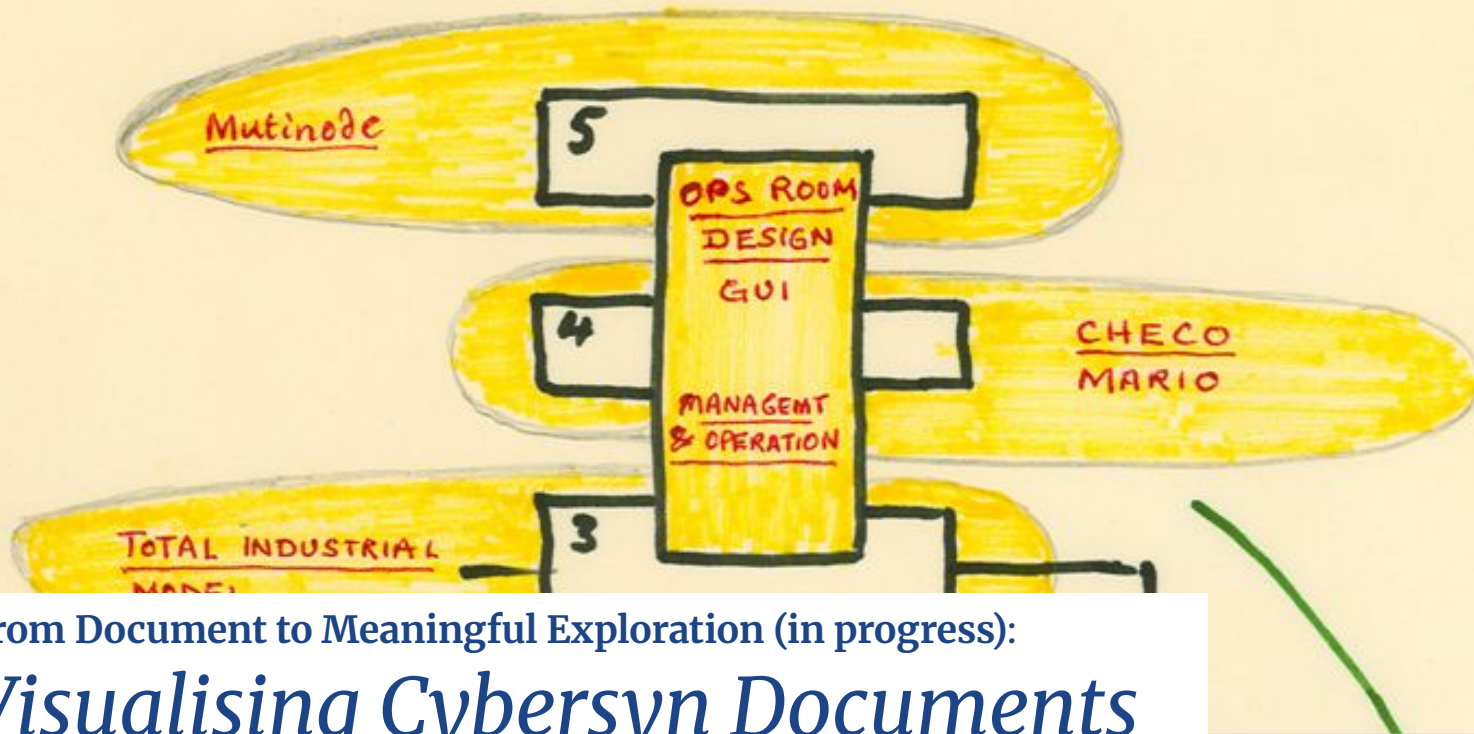




From Document to Meaningful Exploration (in progress): *Visualising Cybersyn Documents*

Ricardo Vega Mora
PhD Student in Computer Science Engineering
Pontificia Universidad Católica de Chile (Professor Juan Reutter)
University of Edinburgh (Professor: Melissa Terras & John Vines)

SB100Y Conference
18, Sept, 2026

Cybersyn Project Images: Courtesy of the Stafford Beer Archive at LJMU Special Collections & Archive.
Digital images courtesy of Constantin Malik.
Other Images: Courtesy of his authors.

https://github.com/ricardovv/ricardov_web_2026/tree/main/pdf

Outline

Personal Relation with Technology

Visualisation Project

Framework

Study Case

Methodology

Project Status / Challenges

Personal Motivation

Personal Relation with Technology

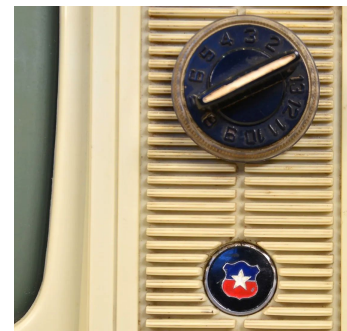
Where I come from...

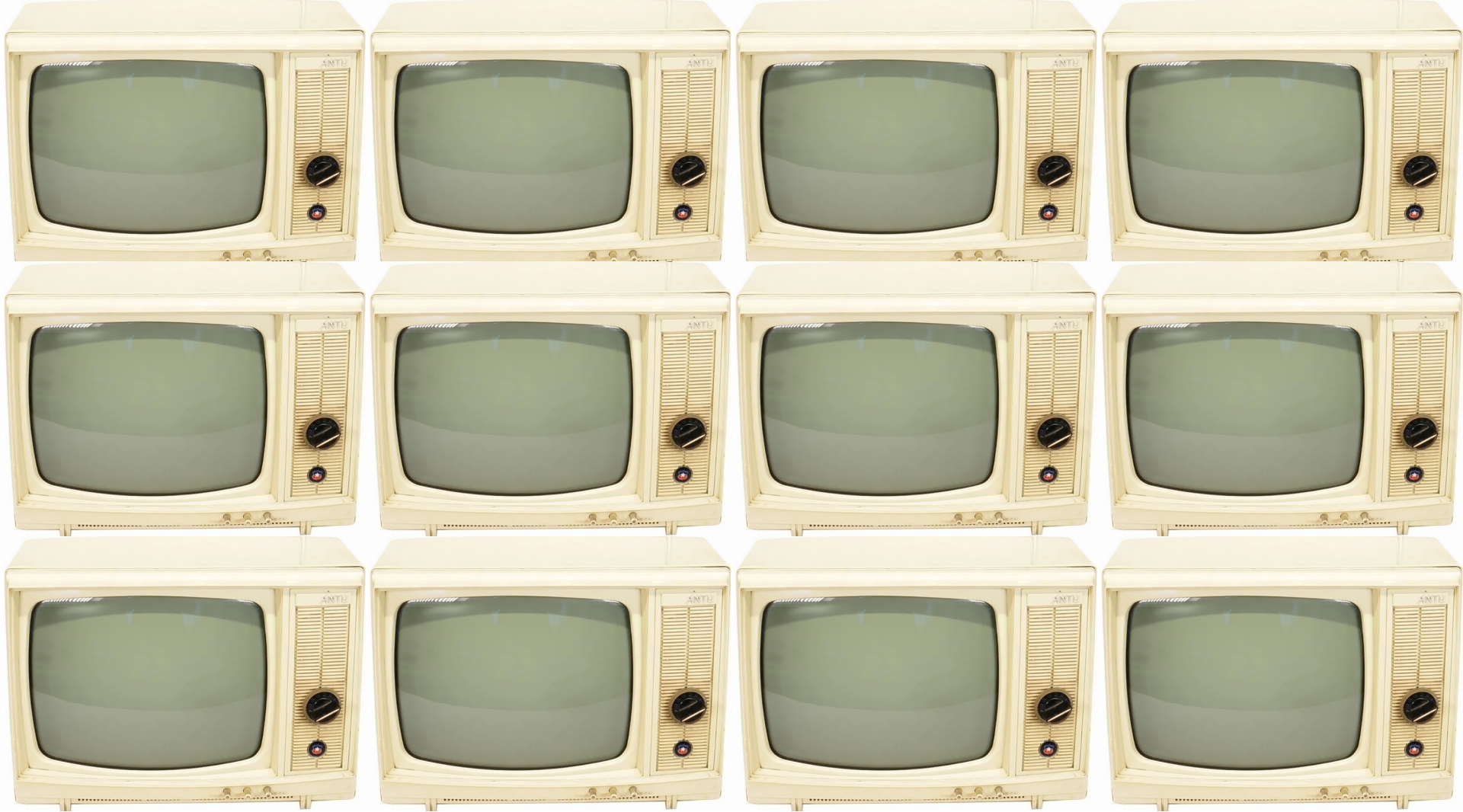


Santiago, Chile.
Designer, artist, media artist.

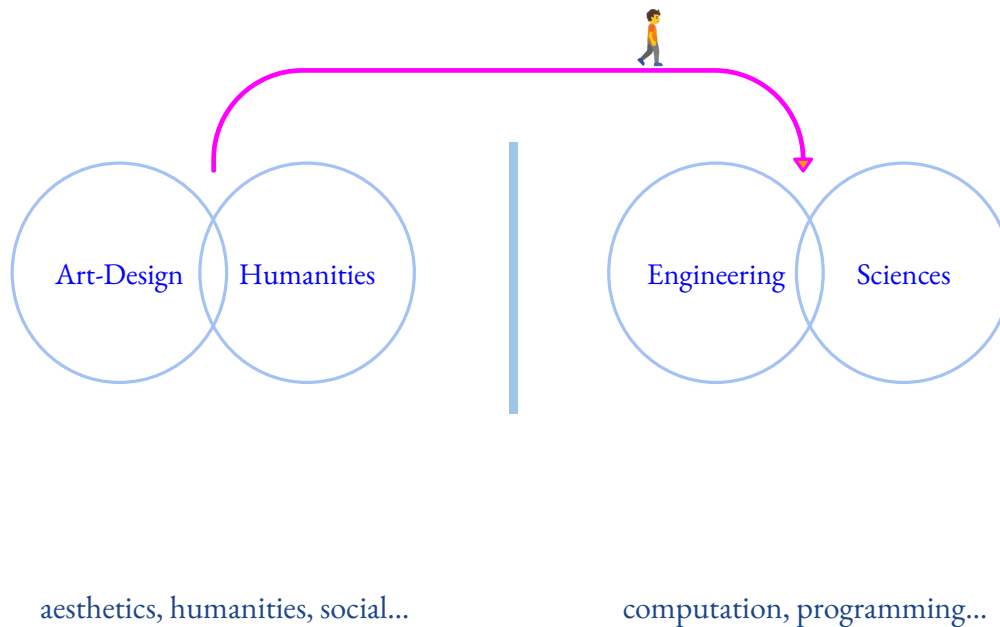
Childhood during the dictatorship, yet surrounded by technology:
televisions (Antu), radios, irons, etc. Electricity.

ANTU TV made in Chile
in the 70's

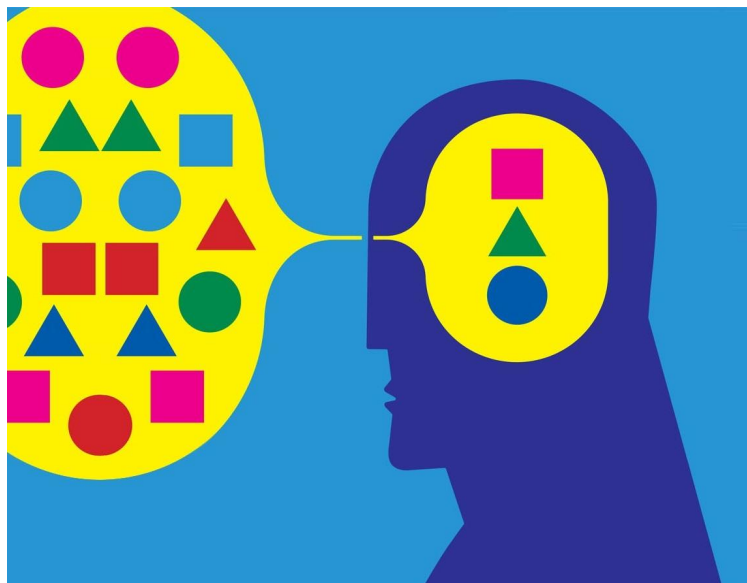




Where I come from...



Interests in technology and image: Visual explorations



[illegible]

Propuesta Constitución 2022

Consolidado normas aprobadas para la propuesta constitucional por el pleno de la Convención

Este documento es un consolidado de las normas aprobadas por la Convención Constitucional, ordenadas por comisión. La relación de las normas aprobadas por la Convención Constitucional, ordenadas por comisión, ya que ese es el proceso de la Convención Constitucional, ordenadas por comisión. La relación de las normas aprobadas por la Convención Constitucional, ordenadas por comisión, ya que ese es el proceso de la Convención Constitucional, ordenadas por comisión.



Cross citation articles analysis

Disciplinary Interests In Cybersyn (Chile)

Movement to recover the history of Latin American technology and Cybernetics:

Dusan Cotoras (sociologist), Rodrigo Fernández (sociologist), David Maulen (art researcher), Hugo Palmarola (designer) & Pedro Alonso (architect), Fernando Portal (architect), Enrique Rivera & Catalina Ossa (artists), Leonardo Portus (artist), Claudio Gutierrez (computing) & Juan Álvarez (computing), Juan Carlos Letelier (biologist), etc.

Some question remains:

What other issues and figures require research?

Arturo Aldunate Philips, Ricardo Uribe, etc.



Chilean Scientist Arturo Aldunate Philips & Norbert Wiener (nd).

Does cybernetics persist in Chile, and under what form?

Biology as unified perspective. Santiago School of Biology



Instituto de filosofía en Ciencias de la complejidad, Dirección Pablo Razeto.

Initial encounters with Cybersyn: Troyano Collective book

Initial encounters with the figure of Stafford Beer:

Article on Cybersyn (Enrique Rivera & Catalina Ossa, artists):

–Colectivo Troyano, Ignacio Nieto, Italo Tello, Ricardo Vega (Art):

Book: Instalning: Art & Technology (2006)



LAS HERRAMIENTAS DE LA CIENCIA PARA EL PUEBLO.

Catalina Ossa y Enrique Rivera

Cuando se planeaba el lanzamiento oficial de Cybersyn a la sociedad chilena, en una fecha ya confirmada para Septiembre de 1973, el Presidente Salvador Allende preparó un especial discurso que pronunciaría en la inauguración:

"Estimados amigos, quiero dedicar la bienvenida personalmente a este lugar, ya que he tenido un interés personal en su desarrollo, y por que quiero pedirles un especial interés en él. Lo que ven alrededor es el resultado de 18 meses de trabajo duro por un grupo sólido de nuestros propios ingenieros chilenos, que se han dedicado a solucionar los problemas de la gestión en las empresas. Ellos han creado para nosotros, una serie de herramientas para ayudarnos en la tarea de controlar la economía.

La ciencia moderna, y en particular la computación electrónica, ofrecen al gobierno una nueva oportunidad para tratar con los complejos problemas modernos de la economía.

Hemos encontrado que en los llamados países avanzados, el poder de la ciencia no ha sido utilizado aún. Hemos desarrollado un sistema con nuestro propio espíritu. Lo que escucharan hoy día es revolucionario - no simplemente por que esto es la primera vez que esto se realiza en el mundo. Es revolucionario por que estamos ante un esfuerzo deliberado para darle a la gente el poder que la ciencia nos da, en una forma en la cual la gente podrá usarla libremente."

Lamentablemente, este discurso nunca pudo ser pronunciado debido a la desestabilización social y económica del Gobierno Socialista, causada por factores internos y externos, que finalmente terminaron con el Golpe Militar de Septiembre de 1973).

Las herramientas a las que se refiere Allende se podrían identificar hoy como los primeros intentos de una emergente realidad sociocultural, que hoy en día son fáciles y utilizadas gracias a diversos avances tecnológicos, y a las posibilidades de interconexión a distancia que permite INTERNET, remarcados en los paradigmas presentes en la era de la Sociedad de la Información y la emergente Sociedad del Conocimiento.

La revolución política, económica, social y tecnológica que representaría Chile estaría materializada por el proyecto CYBERSYN.

101

Disciplinary Interests In Cybersyn

What is the relationship between art and technology in Chile?

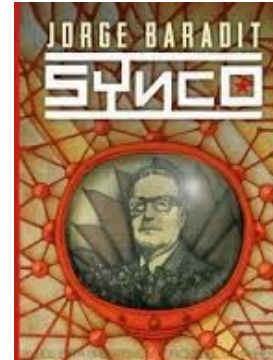
Exploratory art and design projects regarding Cybersyn:



–Enrique Rivera & Catalina Ossa (2002, artists): thesis project at ZKM | Center for Art and Media, Karlsruhe.



–Greg Borenstein & Jem Axelrod (2006, media artists): critical explorations of the interface from a science fiction perspective.



–Jorge Baradit (2006, writer), "SYNCO".



–Mario Navarro (2006, artist): video and installations concerning Stafford Beer "Whisky in OpsRoom"

Disciplinary Interests In Cybersyn

What is the relationship between art and technology?
Exploratory art and design projects regarding Cybersyn:



–Diego Gómez (2014, designer):
archive, documentation, and references



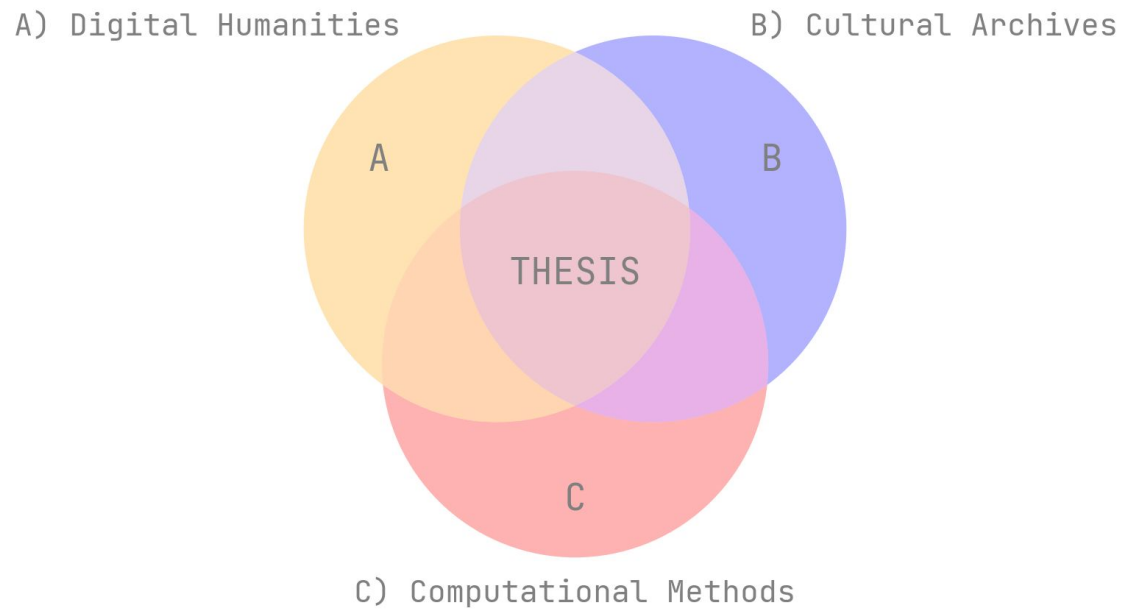
–Luke Munn & Liam Magee (2024,
media & social sciences): Rethink
Economic Futures AI interviews using
LLMs.



–Hugo Palmarola, Pedro Alonso, Edén
Medina (2025, designer, artist,
historian): exhibition and book, real size
OpsRoom.

Visualisation Project

Interests



Questions I had / Needs observed

data processing
& analysis

difficult
complex
time consuming

...

—

data representation
& communication

fit data with graphics
standard graphics not appropriate
time consuming

...

16

The process was slow, heavy, technically difficult for non-experts...

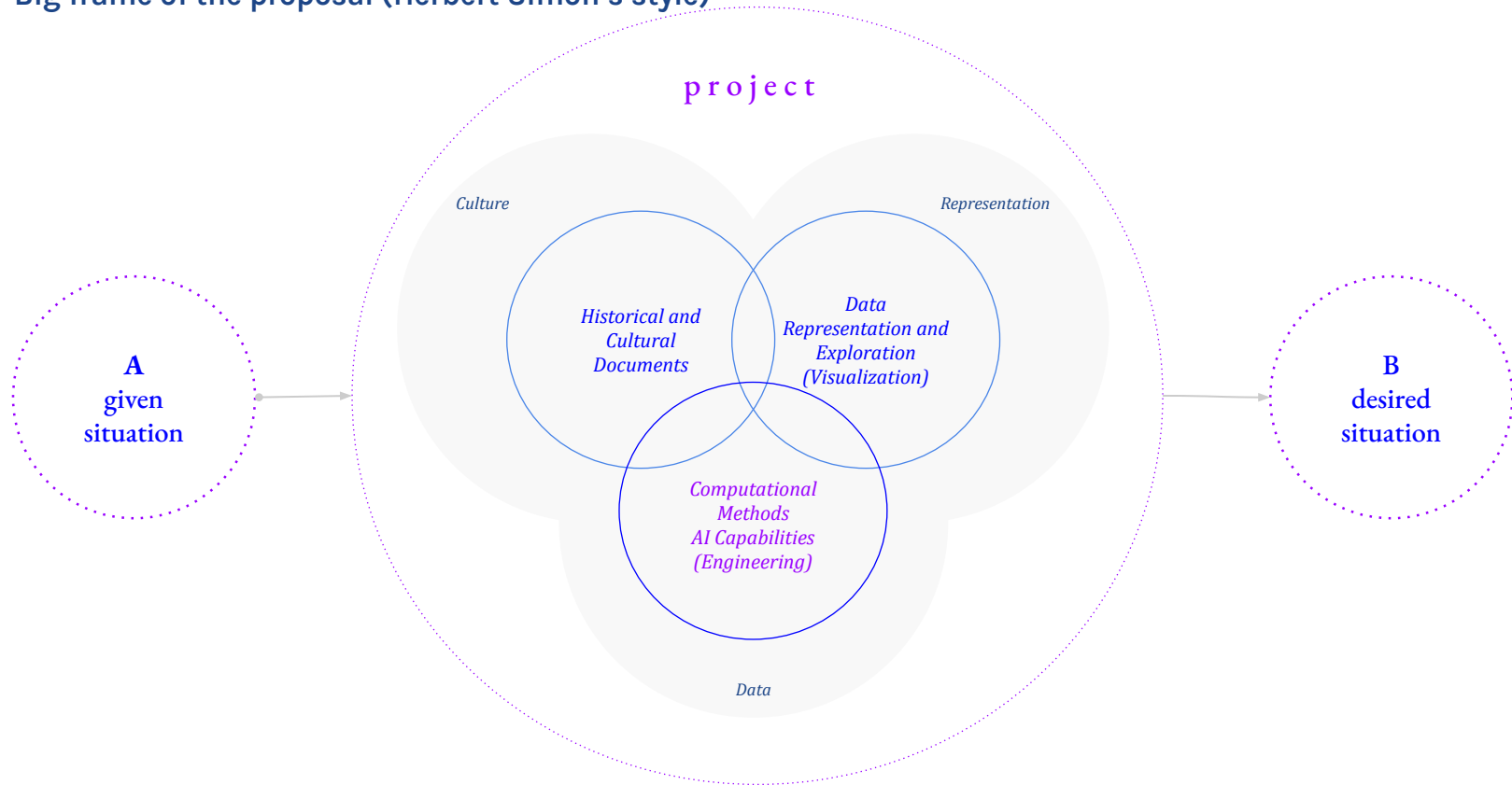
so...

How to optimize tasks within the process?

How to understand and manage the entire process?

(later I changed my questions...)

Big frame of the proposal (Herbert Simon's style)



Proposal

The proposal is to develop an **interactive exploration system** that, through **artificial intelligence-assisted** queries to the database, enables the **generation of interactive visualization narratives** about the relationships and stories of **people, places, or events**.

Relevance & Research Contribution

Visualization of cultural Documents:

This is an opportunity to access the documents and facilitate its visual exploration
(Hinrichs et al., 2015)

Cultural Analytics for the Archive:

The cultural analytics framework will provide another perspective for understanding documents
(Schich et al. 2014; Aiello & Simeone, 2019)

Prototype of Framework for Visualization:

This Framework/Process may serve as a guide for implementing other case studies with multimodal cultural documents.
(Dürsteler & Engelhardt, 2007; Basole et al 2024)

Disciplinary Integration:

This an opportunity for disciplinary integration
(Weaver, 1948)

Recovery of Technological Heritage:

Relevant for understanding current technological developments. Cybersyn is a case study
(Letelier, 2022; Medina, 2014)

Process Overview:

To understand how LLMs and AI expand the creation and analytical possibilities of the visualization process (Dürsteler & Engelhardt, 2007; Basole et al 2024)

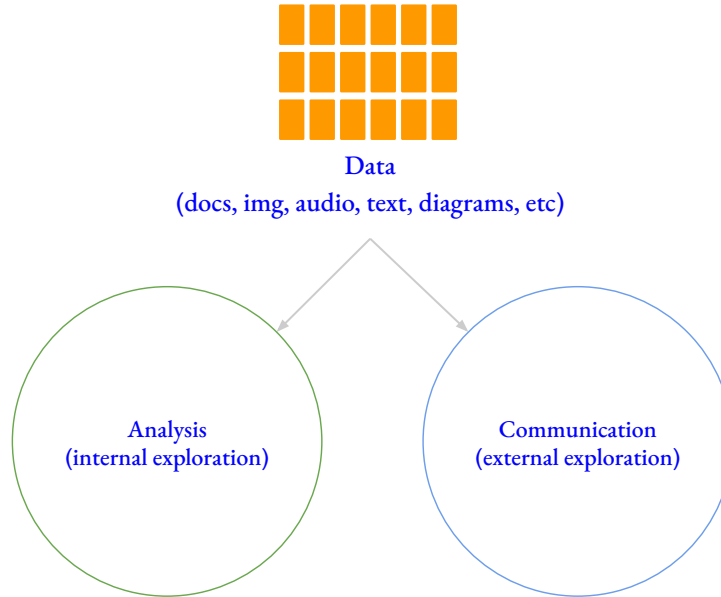
Framework

Visualization

Reduces complexity using visual elements

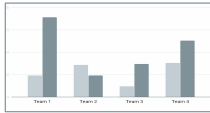


Visualization Main Roles



Visualization

The broader visualization spectrum...



Pragmatic
Visualización



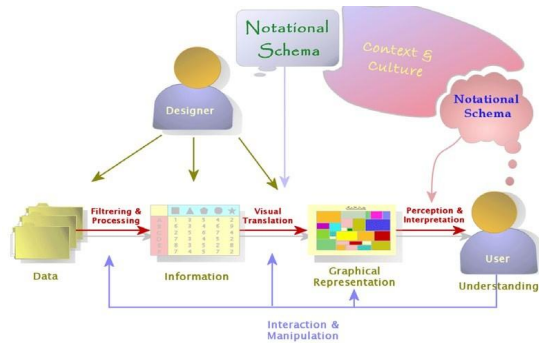
Data Visualization - Exploratory Visualization - Casual Infovis - Physical Visualization - Environmental Visualization - Data Art



Artistic
Visualization

Visualization

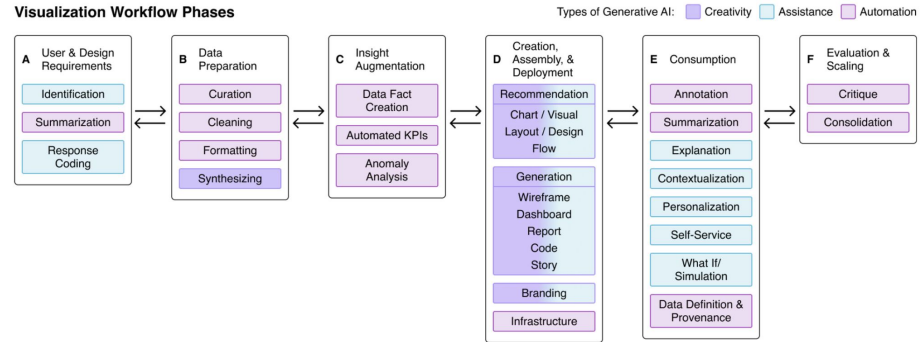
Models: How to understand visualization tasks



"Infovis Diagram":

Most general description of the visualization process, but not considers the AI development.

Visualization Workflow Phases

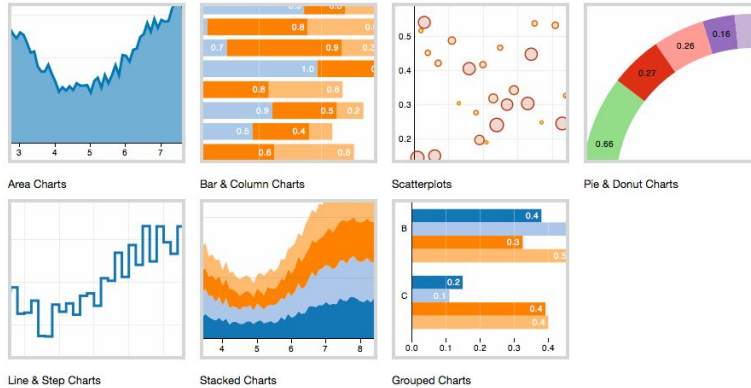


"Visualization Workflow Phases":

Recently discovered, more detailed, focused in find how to integrate AI.

Visualization

Standard graphics vs custom made



Standardized libraries,
Graphics “templates” for each kind of visualization.
Matplotlib, Seaborn (Python)
D3, Protovis (Javascript)
GGPLOT (R)

- limited customization
- mainly for analytics
- + well suited for data
- ...



Custom made visualization
From the ground or based on previous libraries.

- + control over output
- + control over visual style
- + adapted for specific cases
- ...

Visualization

Custom visualization



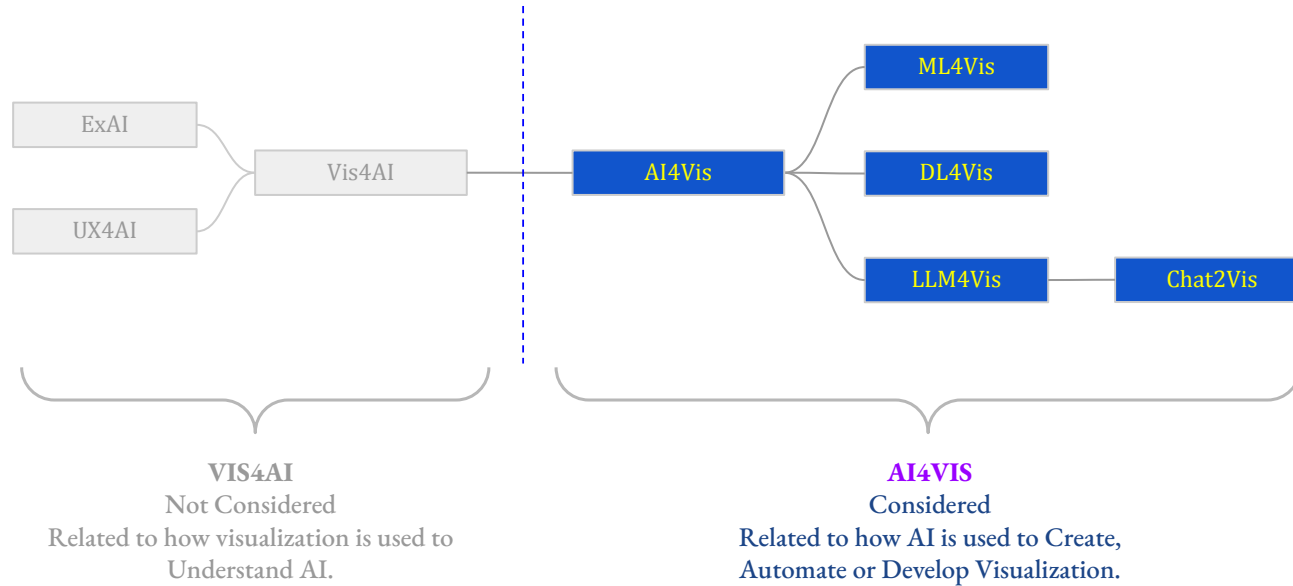
Codex Atlanticus

Interactive website, allows you to navigate through pages, sketches and notes from Leonardo da Vinci Codex Atlanticus.

link: <http://codex-atlanticus.it/#/?lang=EN>

State of the Art

AI and Visualization current research - Two sides



Study Case

Case Study: Stafford Beer Archive, Liverpool John Moores University (LJMU) Documents of relevant project, Chile 1972.

Then



Stafford Beer
English scientist.

Ops Room and usage of
visualizations

Now



Eden Medina Book

Recent render reconstructing
the Ops room.

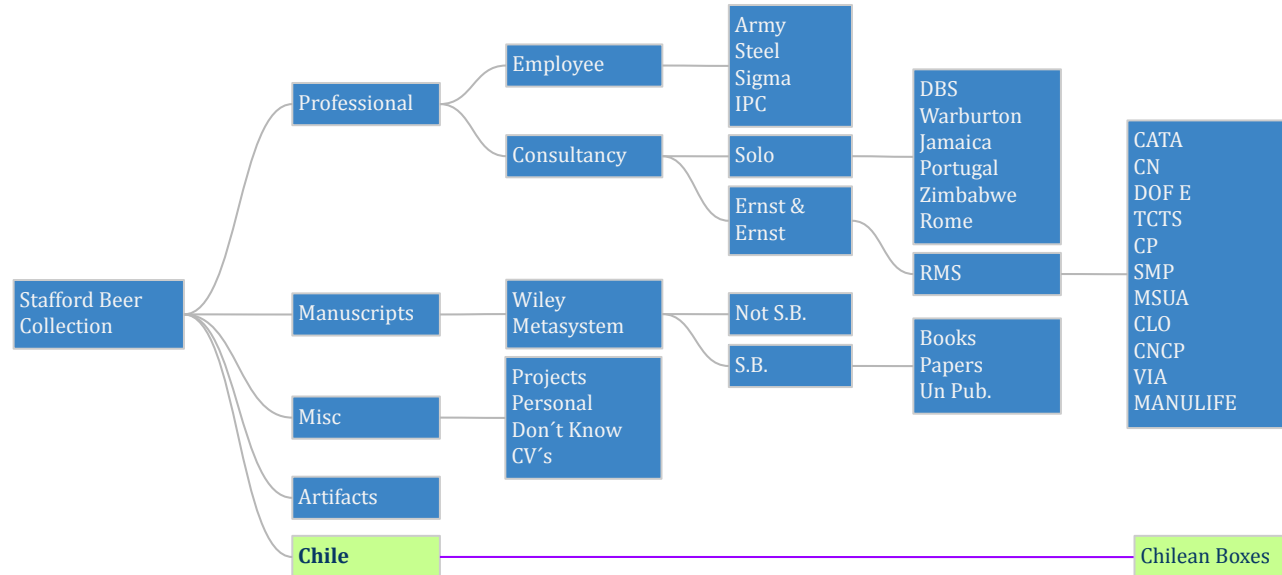
Recent Exhibition
Book

Stafford Beer Archive, Liverpool John Moores University
Special Collections & Archives.

**There is no visual access to explore the
documents on which the research is based.**

Case Study

SBA (LJMU) - Structure



Case Study

SBA (LJMU) - Available Boxes

Stafford Beer
Collection

Chilean Boxes

Box 55 (10 documents): "Hello," and "Please stay". Some items are on loan from this box
Box 56 (3 documents): The Big Black File. This whole box is on loan, and not currently accessible
Box 57 (8 documents): Cyber Stride. Some items are on loan from this box
Box 58 (3 documents): Ron Anderton
Box 59 (1 documents): Arthur Andersen
Box 60 (4 documents): Cybersyn. Some items are on loan from this box
Box 61 (5 documents): Project Cyberfolk. Some items are on loan from this box
Box 62 (15 documents): Ops Room. Some items are on loan from this box
Box 63 (15 documents): "Snapshots". Some items are on loan from this box
Box 64: (15 documents): Corfo Miscellaneous.

Boxes
Available

Some items are on loan from this box. Checked up to here 12/05/2022

Box 65: Corfo Miscellanea 2
Box 66: Correspondence
Box 67: Films, Accounts
Box 69: Chile -POST COUP -Relief
Box 70: POST-COUP Correspondence FOR THE ATTENTION OF SB BEER
Box 71: Related Journals

Boxes Not
Available

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Cybersyn Archives Related to Chile

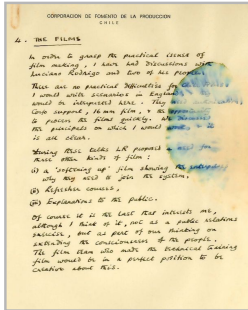


Documentos por cada caja

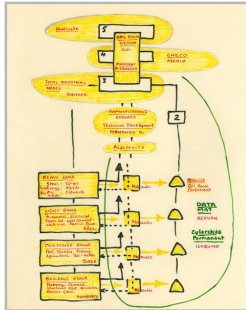
Figure 1 displays a grid of 64 boxes, each containing a 4x4 matrix of numbers. The boxes are labeled Box 55 through Box 64. The numbers are arranged in a grid that is 8 rows by 64 columns. The numbers are arranged in a grid that is 8 rows by 64 columns. The numbers are arranged in a grid that is 8 rows by 64 columns.

Case Study

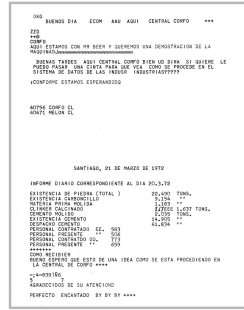
SBA (LJMU) - File Types



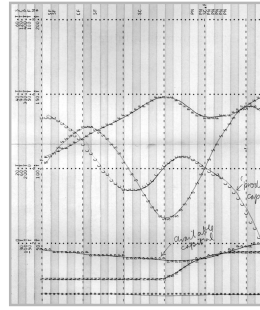
–manuscript (152)



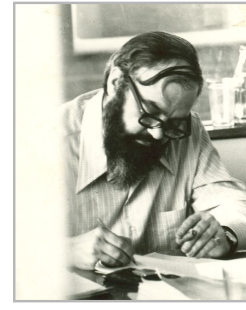
–manuscript diagram
(138)



–official document (74)



–diagram (31)



–photo (15)

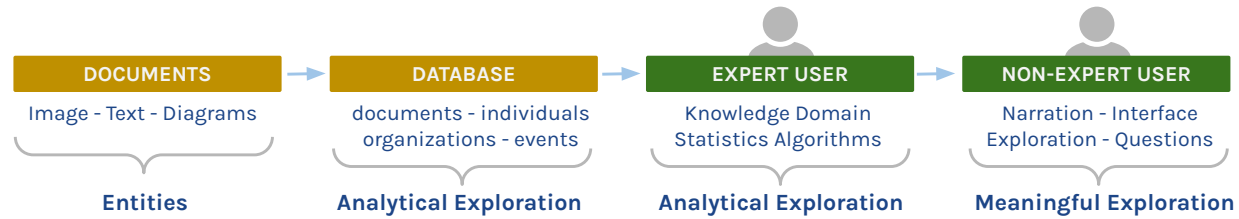


–official letter (8)

Methodology

Methodology

Computational Side

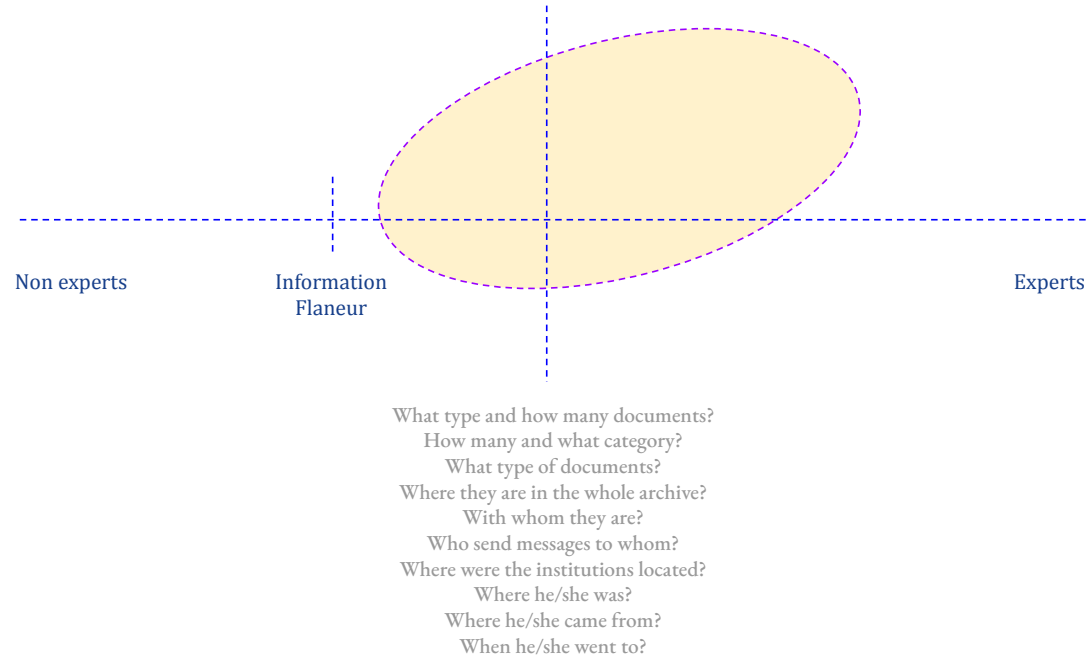


35

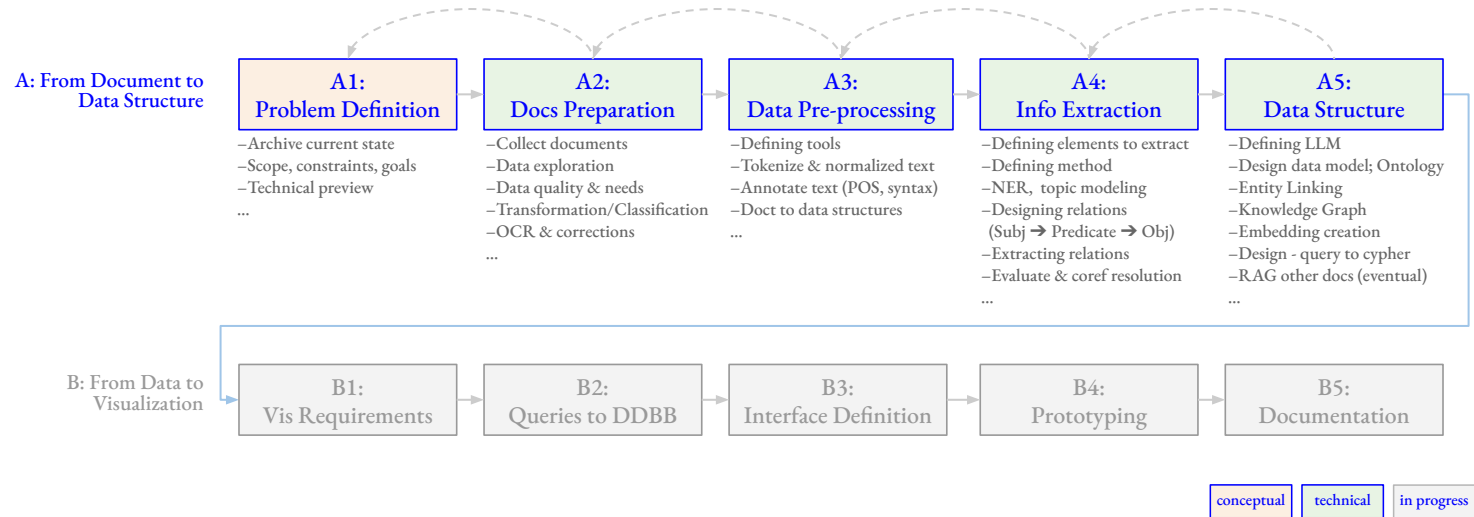
Methodological Side

Mixed Methods: Interviews / Literature / Text Analysis / User Testing ...

Users possible questions



Project Main Tasks & Paths:



37

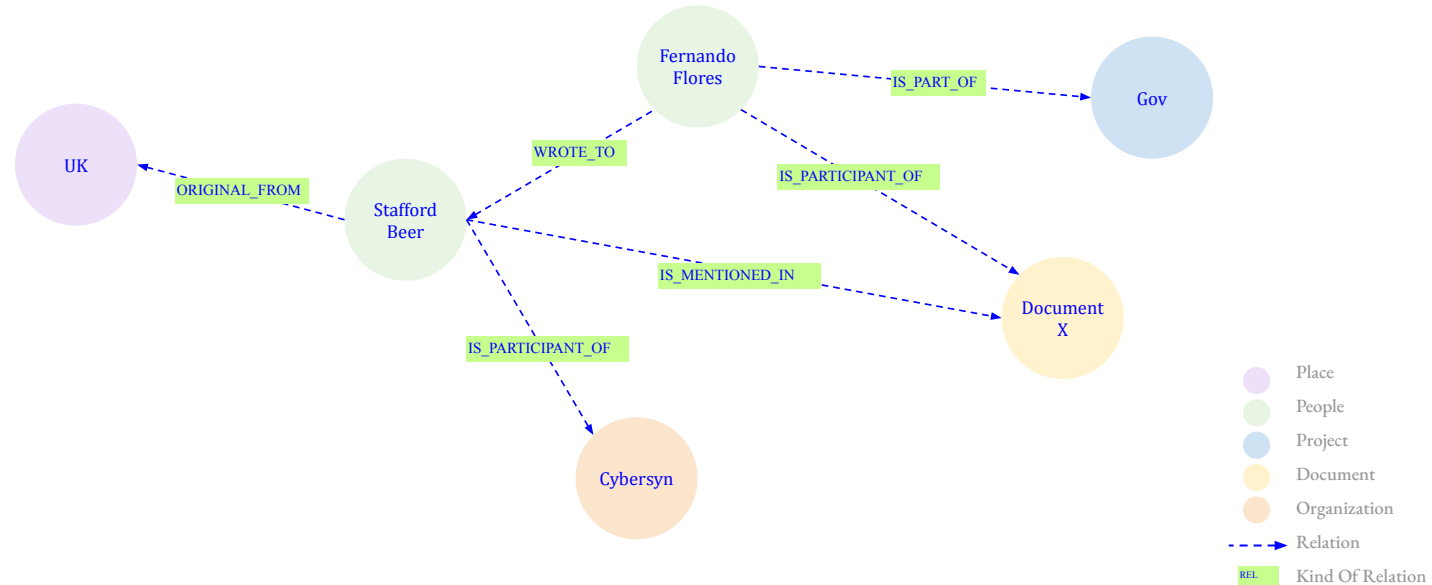
Visualization project
Doing

Auto ethnography & Doc.
Reflecting

Tesis
Writing & Documentation

Knowledge Graphs for DDBB: Why

Triplet “Subject” → “Action” → Objects”
(depending the case)



KGs permit representation of complex relationships among different types of entities, such as people, events, places, concepts...

KGs helps to go from a traditional document-centric approach to a more entity-centric approach.

Competency Questions to Create Entities & Relations

Competency Questions

Factual questions (direct properties of an entity):

- Who invited Stafford Beer to Chile?
- Who was in charge of Cyberstryde?
- What components made up Project SYNCO?

Indirect relationships (intermediate connections, cross-referencing documents, organisations, and people):

- Which documents were discussed in the correspondence between Stafford Beer and the Opsroom team?
- Which letters represent Beer's defence of the project against political attacks?

Structural questions (groupings or temporal evolution):

- Which cybernetic concepts appeared most frequently in the documents during 1972?
- Who worked in the OpsRoom?

Hypothetical or causal questions:

- Which individuals linked the government to factory operations within Project Cybersyn?
 - Which document mentions government officials involved in Project Cybersyn?
- INITIAL ONTOLOGIES

Entities & Relations

~~ENTITIES~~

PERSON
PLACE
ORGANIZATION
EVENT
PROJECT
CONCEPT
TECHNOLOGY
DOCUMENT

~~RELATIONS~~

IS_PARTICIPANT_OF (Actor > Project)
WAS_AT (Actor > Place)
PARTICIPATED_IN (Actor > Project)
WROTE_TO (Actor > Actor)
WORKS_FOR (Actor > Institution)
COLLABORATED_WITH (Actor > Actor)
IS_AUTHOR_OF (Actor > Document)
WAS_MEMBER_OF (Actor > Institution)
IS_RECIPIENT_FROM (Actor > Document)
USED_IN (Technology > Project)
DEVELOPED_BY (Project > Actor)
WAS_USED_IN (Technology > Project)
DEVELOP_CONCEPT (Document > Concept)
SITUATED_IN_PHASE (Document > Operational Phase):
SHARE_TOPIC (Document > Document):
REFERE_TO (Document > Document)
MENTION_TO (Document > Actor)
TREAT_ABOUT (Document > TopicConcept)
IS_PART_OF (Document > CollectionSection)
THIS_LOCATED_IN (Document > Geographic Location):
TRAVELED_TO (Actor > Place)
ORIGINAL_FROM (Actor > Place)
OCCURS_IN (Document > TimePeriodInstant)

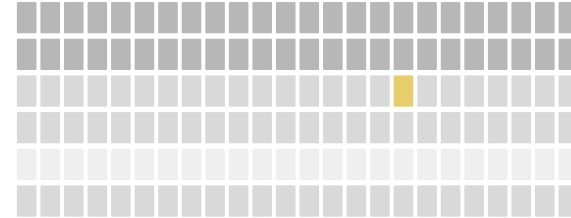
Visualization levels & scopes

Distribution

What type and how many documents?
How many and what category?
...



Macro - whole archive
structure, categories,
clusters



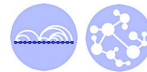
Part-to-a-whole - Proportions

What type of documents?
Where they are in the whole archive?
...



Relationships

With whom they are
who send messages to whom else?
...



Meso - categories/clusters
documents, persons,
institutions
relations, positions, etc.



Location

Where were the institutions
located?
Where he/she was?
Where he/she came from?
When he/she went to?
...

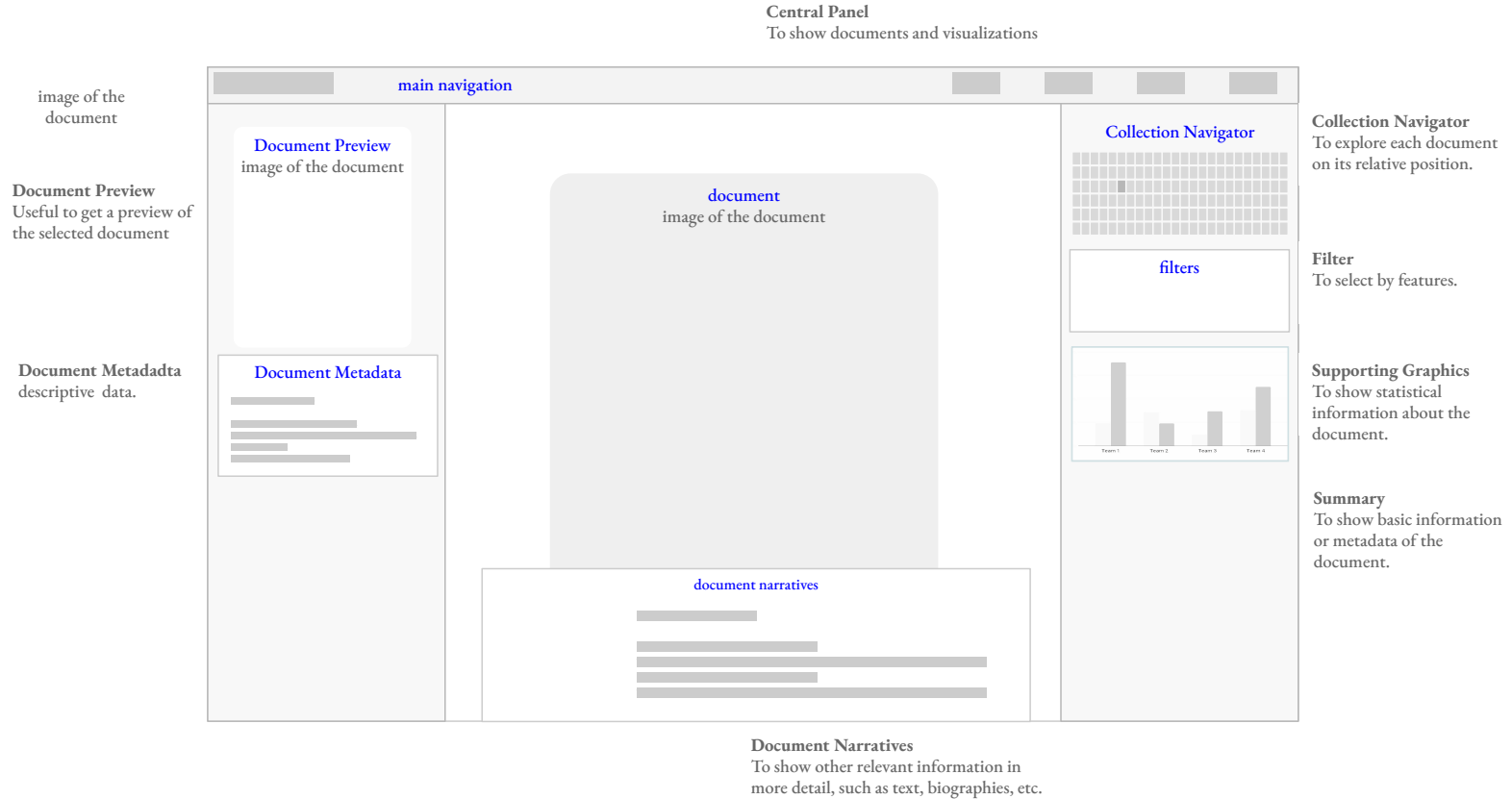


Micro - single entity...
documents, persons,
institutions relations,
positions, etc.



Project Proposal

Possible Layout for elements (provisional)



Project Status

Prototyping text extraction from image (Tesseract)



3rd August 1873

Dear Raul

It has taken some time for me to deliberate the enclosed letter to the President.

I have said in it only those things which my professional conscience dictates must be said.

Therefore I am relying on you to explain the necessity of the letter to Fernando, since after all a statement was demanded of me by the President ; and then to transmit the letter with an appropriate translation.

Yours ever

Prototyping image description (GPT, Gemini)



Esta es una fotografía histórica en blanco y negro que captura a un grupo de figuras clave relacionadas con el Proyecto Cybersyn en Chile.

A continuación, describo los elementos principales de la imagen:

Sujetos y Contexto Histórico

Personajes: De izquierda a derecha, se observa a Gui Bonsiepe (diseñador industrial alemán), seguido por Stafford Beer (el cibernético británico, fumando), Herman Schwember y Fernando Flores (ministro de economía y pieza central del proyecto).

Ubicación: El pie de foto indica que la escena tiene lugar en el Restaurante Cura Nurin, en Viña del Mar (Chile).

Fecha: La imagen está datada en 1971, durante el gobierno de Salvador Allende.

Composición y Estilo

Acción: Los cuatro hombres aparecen caminando al aire libre en lo que parece ser una conversación informal. La actitud es relajada; Stafford Beer sostiene lo que parece ser un cigarrillo o puro cerca de su boca.

Vestimenta: Refleja la época y un estilo intelectual/informal. Se observan cardiganes de punto, chaquetas ligeras, camisas abotonadas y anteojos de pasta gruesa.

Entorno: Al fondo se aprecia un paisaje con vegetación, una estructura de piedra (posiblemente un muro o entrada) y una casa de estilo clásico sobre una colina. También se ve un vehículo antiguo estacionado a la izquierda.

Prototyping entity extraction (GPT, Gemini).



```
img_text = """
3rd August 1873
```

```
Dear Raúl
```

```
It has taken some time for me to
deliberate the enclosed letter to
the President.
```

```
I have said in it only those things
which my professional conscience
dictates must be said.
```

```
Therefore I am relying on you to
explain the necessity of the letter
to Fernando, since after all a
statement was demanded of me by the
President ; and then to transmit
the letter with an appropriate
translation.
```

```
Yours ever
"""
```

```
prompt_img = """From the text provided between accents\
contains or deals with a photo description, you will identify
the following:
```

1. Identify if it is a photo description (true boolean - is_description)
2. Identify the people mentioned (people)
3. Identify the places mentioned (places)
4. Identify the organizations mentioned (organization)
5. Identify dates (dates)
6. Extract 3 keywords, 1 word per keyword (keywords)

```
If it does not contain people, write "no people"
If it does not contain places, write "no places"
If it does not contain dates, write "no dates"
If it does not contain organization, write "no organization"
```

```
Deliver the response as a JSON object with the following names:
is_description:, people:, places:, dates:, organization:,
Keywords:,
```

```
If the provided text does NOT contain or DOES NOT It's about an
photo, simply write "It's not about photo."
```

```
```{text_to_json}```
"""
```

```
resp_prompt_ent1 = obtener_respuesta(prompt5_ent1.format(
mi_textito = img_text))
```

```
print(resp_prompt_ent1)
```

```
Título: No hay título
Personas: Raúl, Fernando
Lugares: No hay lugares
Fechas: 3rd August 1873
```

## Challenges

*Proposed program for developing Dyn. Mod. of Economy*  
 PROGRAMA PROPUESTO PARA DESARROLLAR EL MODELO DINAMICO DE LA ECONOMIA.

1. Estudio exhaustivo del Checo 1.  
 Implicaría: *study in detail*  
 1.1 Estudiar a fondo la Teoría de la Dinámica Industrial. *study in detail*  
 1.2 Estudiar a fondo el Modelo Económico planteado. *practice with*  
 1.3 Practicar con el CHECO 1, con el fin de adquirir familiaridad en el manejo e Interpretación de las Variables de Proceso, Parámetros y Resultados, así como una comprensión de la Dinámica del Proceso en General.  
 1.4 Determinar con mayor precisión las Constantes, Variables y parámetros que resulten de mayor relevancia en el Modelo. *Practice with better precision*  
 1.5 Preparación de un Equipo destinado a abordar el Problema Computacional involucrado. *Release a Team*
2. Modificaciones al Checo 1. *searching for a better Model*  
 Una vez adquirido el dominio del CHECO 1, la etapa siguiente consistiría en evolucionarlo hasta un Modelo que represente mejor nuestra Economía. Implicaría:  
 2.1 Replanteamiento del CHECO 1 a la luz de la sensibilidad de los Factores e Interacciones mas significativos. *Reserve in mind possibility of interaction*  
 2.2 Análisis de aquellas Areas del Modeló que requieren de una mayor precisión. Considerando la posibilidad de desarrollar Modelos Complementarios para dichas Areas. *with complementary Models*  
 2.3 Experimentación con el Modelo Modificado.  
 2.4 Pronunciamiento del Equipo de Computación sobre la Naturaleza del Compilador requerido y de la Problemática asociada. *Compilation Team decides on compiler*
3. Transferencia de Software.  
 Implicaría:  
 3.1 Adquirir el Compilador adecuado para la utilización efectiva de la Dinámica Industrial.  
*Buying Compiler*

CORPORACION DE FOMENTO DE LA PRODUCCION  
 CHILE

### THE ROOM

CORFO to provide plans of the room in all nine dimensions.

The Operations Room should be thought of  
 NOT as a room containing interesting bits of equipment  
 BUT as a control machine comprising men and artifacts in symbiotic relationship

It needs designing as a totality and as an operational entity.

This is a problem in ergonomics.

A three-dimensional model of the room will be gradually constructed in London with the help of Gui Bonsieppe in Chile.

We will keep in touch by Telsax and especially by exchanging photographs.

Special attention will be paid to the development of man-machine interfaces, using 'big hand' controls.

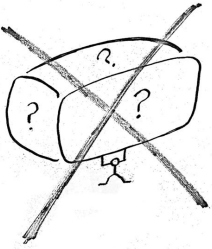
# Challenges

3 First Principle : GOVERNMENT IS THE PEOPLE'S HELP

**FINISHED**

The view of government

- as incomprehensible
- as the people's burden



STOP THIS ↑ WHENEVER YOU CAN

Nunca despara

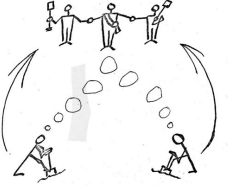
CHANGE is a state of mind that everyone shares

4

**STARTED**

The view of government

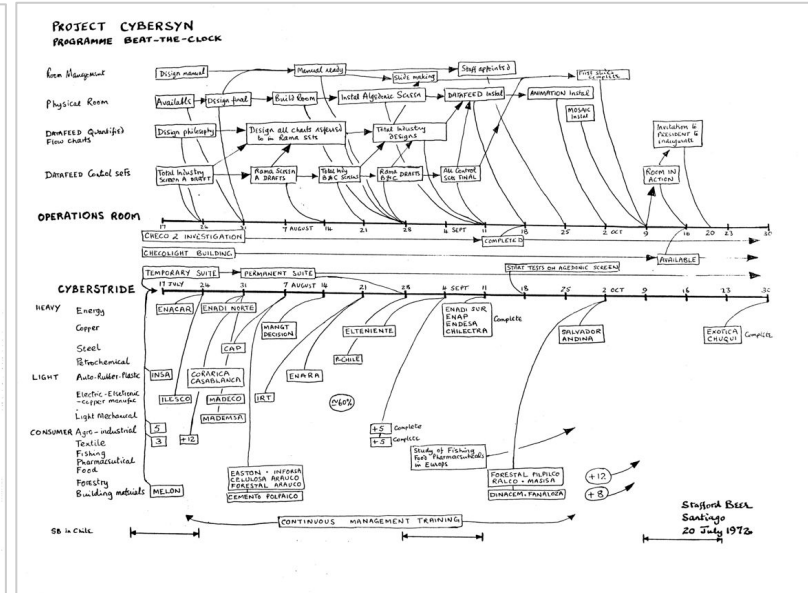
- as the people thinking what to do
- as acting together to do it

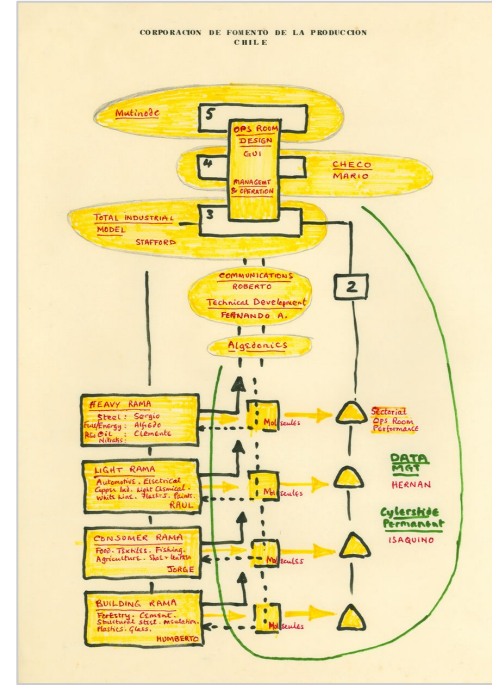
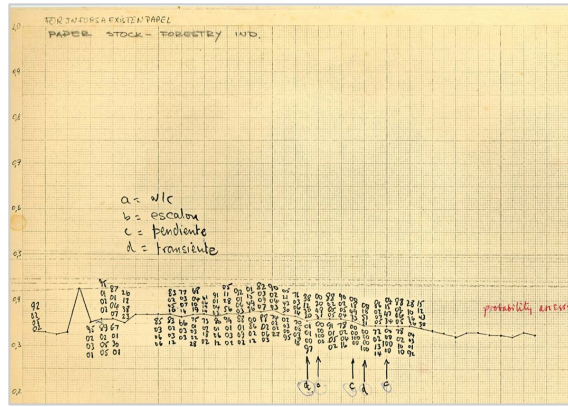
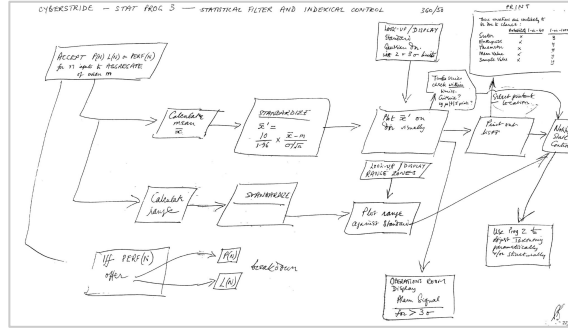


The Goparito President says : "Government is the people".

The wishes of the people will be made known to the Government at all times.

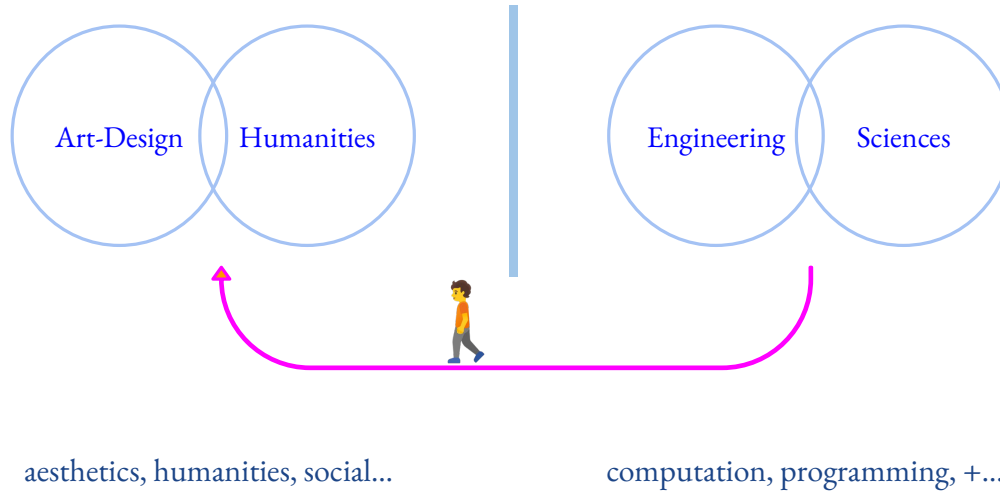
We share the TECHNOLOGY, which belongs to the people, to do it.



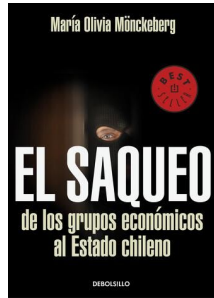
[illegible]

# *Personal Motivation*

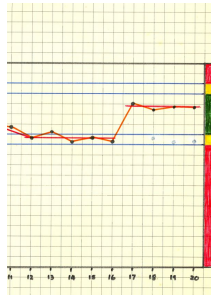
## What to do with this knowledge & experience



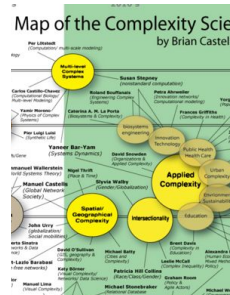
## Motivation / Projections



**Book Visualization**  
"El saqueo de los grupos económicos al estado de Chile", 2015. María Olivia Mönckeberg.



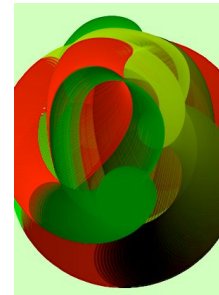
**Cultural Documents**  
Reconstructing the diagrams of the Cybersyn project



**Scientific Information**  
Castellani, B. (2019). Map of the complexity sciences. [art-sciencefactory.com/complexity-map\\_feb09.html](http://art-sciencefactory.com/complexity-map_feb09.html)



**Forensic Visualization**  
DDHH related issues.



**Computational art**  
Personal

**Propuesta Constitución 2022**  
*Consolidado normas aprobadas para la pleno de la Convención*

Este documento es un borrador por consultar. La redacción definitiva, ya que es preceptiva.

CAPÍTULO PRIMERO		CAPÍTULO SEGUNDO		CAPÍTULO TERCERO		CAPÍTULO CUARTO		CAPÍTULO QUINTO	
ARTÍCULO	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9
10	10	10	10	10	10	10	10	10	10
11	11	11	11	11	11	11	11	11	11
12	12	12	12	12	12	12	12	12	12
13	13	13	13	13	13	13	13	13	13
14	14	14	14	14	14	14	14	14	14
15	15	15	15	15	15	15	15	15	15
16	16	16	16	16	16	16	16	16	16
17	17	17	17	17	17	17	17	17	17
18	18	18	18	18	18	18	18	18	18
19	19	19	19	19	19	19	19	19	19
20	20	20	20	20	20	20	20	20	20
21	21	21	21	21	21	21	21	21	21

**Public Information**  
Adding more constitutions:  
Chile 1925, 1980, 2022.  
France, USA, Germany (1948)  
etc...

## Motivation

# Visualization is a tool to challenge *Information asymmetries*

*in*

public information  
scientific information  
cultural information  
humanities

...

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Thank You

Comments & Questions

From Document to Meaningful Exploration (in progress):

# *Visualising Cybersyn Documents*

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University of Edinburgh (Professor: Melissa Terras & John Vines)

SB100Y Conference  
18, Sept, 2026